# Exercise on Basic Finance (try to use excel)-Answers

Question 1:

a. What is the PV of 1000 NOK that will be received in 1 year if the interest rate is 10%?

b. What is the PV of 1000 NOK that will be received in 2 years if the interest rate is 10%?

c. What can you conclude on the relationship between PV and the timing of the cash flow?

d. What is the PV of 1000 NOK that will be received in 1 year if the interest rate is 8%?

e. What can you conclude on the relationship between PV and the interest rate?

**Answer:**

a. PV=1000/1.1=909.1

b. PV=1000/$(1.1)^2$=826.4

c. The further is the time of payment, the lower is the PV.

d. PV=1000/1.08=925.9

e. The lower the interest rate is, the higher is the PV and vice versa.

Question 2:

a. What is the FV after two years of the following cash flow: 1000 NOK at the end of each of the two years if the interest rate is 10%?

b. What is the FV after two years of the following cash flow: 1000 NOK at the beginning of each of the two years if the interest rate is 10%?

**Answer:**

a. FV=1000*(1.1)+1000=2100

b. FV=1000*$(1.1)^2$+1000*(1.1)=2310

Question 3:

You have 1000 NOK that you are willing to invest in the saving program. What would be the best savings option:

1. 1000 NOK that yield 10% in the first two years, 8% in the third and the fourth years and 5% in the fifth year.

2. 1000 NOK that yield 7% in each of the five years.

3. 1000 NOK that yield 5% in the first year, 7% in the second year and 9% in the last three years.

**Answer:**

1. $FV=1000*(1.1)^2*(1.08)^2*(1.05)=1482$

2. $FV=1000*(1.07)^5=1403$

3. $FV=1000*(1.05)*(1.07)*(1.09)^3=1455$

Option 1 is the best option since it yields the highest FV.

Question 4:

You were offered to invest 800,000 NOK in the following business that will yield 100,000 NOK at the end of each of the following 10 years if the annual interest rate is 5%. Will you accept the offer?

**Answer:**

Value of each asset is the PV of its cash flow. The PV of this business is 772,173 NOK. (In Excel: =PV(0.05;10;-100000) — note the payment must be entered as a negative number for Excel's sign convention to return a positive present value). Therefore, you should reject the offer.